Binomial Theorem

Short Answer Questions - Detailed Solutions

1. Expand $(x+2)^5$ using Binomial Theorem.
Solution:
The binomial expansion of $(a+b)^n$ is given by:
$$(a+b)^n = \sum_{r=0}^{n} {}_nC_r \, a^{n-r} b^r$$
Here, $a = x$, $b = 2$, and $n = 5$.
Therefore,
$$(x+2)^5 = \sum_{r=0}^{5} {}_5C_r \, x^{5-r} \cdot 2^r$$
Writing out terms:
$$= {}_5C_0 x^5 \cdot 2^0 + {}_5C_1 x^4 \cdot 2^1 + {}_5C_2 x^3 \cdot 2^2 + {}_5C_3 x^2 \cdot 2^3 + {}_5C_4 x \cdot 2^4 + {}_5C_5 \cdot 2^5$$
Using binomial coefficients:
$${}_5C_0 = 1, \quad {}_5C_1 = 5, \quad {}_5C_2 = 10, \quad {}_5C_3 = 10, \quad {}_5C_4 = 5, \quad {}_5C_5 = 1$$
Substitute and calculate powers of 2:
$$= x^5 + 5x^4(2) + 10x^3(4) + 10x^2(8) + 5x(16) + 1(32)$$
$$= x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$$
So, the expanded form is:
$$\boxed{x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32}$$

2. Find the coefficient of x^4 in the expansion of $(2x-3)^6$.
Solution:
The general term in expansion of $(a+b)^n$ is:
$$T_{r+1} = {}_nC_r a^{n-r} b^r$$
Here, $a = 2x$, $b = -3$, $n = 6$.
So,
$$T_{r+1} = {}_6C_r (2x)^{6-r} (-3)^r = {}_6C_r 2^{6-r} (-3)^r x^{6-r}$$
We want the term where power of x is 4, so:
$$6 - r = 4 \implies r = 2$$
Substitute $r = 2$:
$$\text{Coefficient} = {}_6C_2 \cdot 2^4 \cdot (-3)^2$$
Calculate:
$${}_6C_2 = \frac{6 \times 5}{2 \times 1} = 15$$
$$2^4 = 16, \quad (-3)^2 = 9$$
Multiply:
$$15 \times 16 \times 9 = 2160$$
Thus, the coefficient of x^4 is:
$$\boxed{2160}$$

3. Find the middle term in the expansion of $\left(x + \frac{1}{x}\right)^8$.
Solution:

Number of terms $= n + 1 = 8 + 1 = 9$ (odd number), so middle term is the $\frac{9+1}{2} = 5$th term, i.e., T_5.

General term:

$$T_{r+1} = {}_8C_r x^{8-r} \left(\frac{1}{x}\right)^r = {}_8C_r x^{8-2r}$$

For $r = 4$ (since T_5), power of x:

$$8 - 2 \times 4 = 8 - 8 = 0$$

So the middle term is independent of x.

Compute the coefficient:

$${}_8C_4 = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = 70$$

Therefore, middle term is:

$$\boxed{70}$$

4. Find the term independent of x in the expansion of $\left(x^2 + \frac{1}{x}\right)^9$.
Solution:

The general term is:

$$T_{r+1} = {}_9C_r (x^2)^{9-r} \left(\frac{1}{x}\right)^r = {}_9C_r x^{2(9-r)-r} = {}_9C_r x^{18-3r}$$

For term independent of x, power of x is zero:

$$18 - 3r = 0 \implies r = 6$$

Calculate the coefficient:

$${}_9C_6 = {}_9C_3 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

Hence, the term independent of x is:

$$\boxed{84}$$

5. In the expansion of $(1 - 2x)^7$, find the coefficient of x^3.
Solution:

General term:

$$T_{r+1} = {}_7C_r (1)^{7-r} (-2x)^r = {}_7C_r (-2)^r x^r$$

For x^3, $r = 3$.

Compute:

$${}_7C_3 = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$$

$$(-2)^3 = -8$$

Coefficient:

$$35 \times (-8) = -280$$

So, the coefficient of x^3 is:

$$\boxed{-280}$$

6. Find the 5th term in the expansion of $(2x-1)^6$.
Solution:
The kth term is:
$$T_k = {}_6C_{k-1}(2x)^{6-(k-1)}(-1)^{k-1}$$
For the 5th term, $k=5 \implies r=k-1=4$.
Substitute:
$$T_5 = {}_6C_4(2x)^2(-1)^4 = 15\times(2x)^2\times 1 = 15\times 4x^2 = 60x^2$$
So, the 5th term is:
$$\boxed{60x^2}$$

7. Find the general term in the expansion of $\left(x-\frac{3}{x}\right)^7$.
Solution:
General term T_{r+1} is:
$${}_7C_r\, x^{7-r}\left(-\frac{3}{x}\right)^r = {}_7C_r(-3)^r x^{7-r-r} = {}_7C_r(-3)^r x^{7-2r}$$
Therefore,
$$\boxed{T_{r+1} = {}_7C_r(-3)^r x^{7-2r}}$$

8. Find the coefficient of x^5 in the expansion of $\left(x^2-\frac{1}{x}\right)^6$.
Solution:
General term:
$$T_{r+1} = {}_6C_r(x^2)^{6-r}\left(-\frac{1}{x}\right)^r = {}_6C_r(-1)^r x^{2(6-r)-r} = {}_6C_r(-1)^r x^{12-3r}$$
Set exponent equal to 5:
$$12-3r=5 \implies 3r=7 \implies r=\frac{7}{3}$$
Not an integer, so no term contains x^5.
$$\boxed{0}$$

9. Find the middle term of the expansion of $(1+x)^9$.
Solution:
Number of terms $= 9+1 = 10$ (even).
So, middle terms are 5th and 6th terms:
$$T_5 = {}_9C_4x^4 = 126x^4, \quad T_6 = {}_9C_5x^5 = 126x^5$$
So the middle terms are:
$$\boxed{126x^4 \quad \text{and} \quad 126x^5}$$

10. In the expansion of $(a+bx)^n$, find the term containing x^r.
Solution:
General term:
$$T_{r+1} = {}_nC_r a^{n-r}(bx)^r = {}_nC_r a^{n-r}b^r x^r$$
Thus, the term containing x^r is:
$$\boxed{{}_nC_r a^{n-r}b^r x^r}$$

Long Answer Questions - Detailed Solutions

11. Prove that the sum of coefficients in $(1+x)^n$ is 2^n.
Solution:
The sum of coefficients is the result of substituting $x=1$ in the expansion:
$$(1+1)^n = 2^n$$
This is because:
$$(1+x)^n = \sum_{r=0}^{n} {}_nC_r x^r$$
Putting $x=1$:
$$\sum_{r=0}^{n} {}_nC_r = 2^n$$

12. If the 5th and 6th terms in the expansion of $(1+x)^n$ are equal, find n.
Solution:
The kth term is:
$$T_k = {}_nC_{k-1}x^{k-1}$$
Given:
$$T_5 = T_6 \implies {}_nC_4 = {}_nC_5$$
We know:
$${}_nC_r = {}_nC_{n-r}$$
So,
$$4 = n-5 \implies n = 9$$

13. Show that the middle term of $(1+x)^{2n}$ is ${}_{2n}C_n x^n$.
Solution:
Number of terms in expansion:
$$2n+1$$
Middle term is the $(n+1)^{th}$ term:
$$T_{n+1} = {}_{2n}C_n x^n$$

14. Find the term independent of x in the expansion of $\left(x^2+\frac{1}{x}\right)^9$.
Solution:
Already solved above, term independent of x occurs when $r=6$, coefficient is:
$${}_9C_6 = 84$$

15. Find the term containing x^3 in the expansion of $(2x^2-3x)^6$.
Solution:
General term:
$$T_{r+1} = {}_6C_r(2x^2)^{6-r}(-3x)^r = {}_6C_r 2^{6-r}(-3)^r x^{2(6-r)+r} = {}_6C_r 2^{6-r}(-3)^r x^{12-2r+r} = {}_6C_r 2^{6-r}(-3)^r x^{12-r}$$
Set power of x to 3:
$$12-r=3 \implies r=9$$
$r=9$ is invalid (max $r=6$).
Hence, no term containing x^3 exists.

16. Find the value of ${}_{10}C_4 + {}_{10}C_5 + {}_{10}C_6$.
Solution:
Calculate each:
$$ {}_{10}C_4 = 210, \quad {}_{10}C_5 = 252, \quad {}_{10}C_6 = 210 $$
Sum:
$$ 210 + 252 + 210 = 672 $$

17. Prove that $\sum_{r=0}^{n} {}_{n}C_r = 2^n$.
Solution:
By Binomial Theorem:
$$ (1+1)^n = \sum_{r=0}^{n} {}_{n}C_r 1^r = \sum_{r=0}^{n} {}_{n}C_r = 2^n $$

18. Expand $(1-x)^5$ and verify for $x = \frac{1}{2}$.
Solution:
Expand:
$$ (1-x)^5 = \sum_{r=0}^{5} {}_{5}C_r (-1)^r x^r = 1 - 5x + 10x^2 - 10x^3 + 5x^4 - x^5 $$
Substitute $x = \frac{1}{2}$:
$$ 1 - 5 \times \frac{1}{2} + 10 \times \left(\frac{1}{2}\right)^2 - 10 \times \left(\frac{1}{2}\right)^3 + 5 \times \left(\frac{1}{2}\right)^4 - \left(\frac{1}{2}\right)^5 $$
Calculate each term:
$$ 1 - \frac{5}{2} + 10 \times \frac{1}{4} - 10 \times \frac{1}{8} + 5 \times \frac{1}{16} - \frac{1}{32} $$
$$ = 1 - 2.5 + 2.5 - 1.25 + 0.3125 - 0.03125 $$
Sum:
$$ 1 - 2.5 + 2.5 = 1 \quad \Rightarrow \quad 1 - 1.25 + 0.3125 - 0.03125 = 0.03125 = \frac{1}{32} $$
Also,
$$ (1 - \frac{1}{2})^5 = \left(\frac{1}{2}\right)^5 = \frac{1}{32} $$
Verified.

19. Using Binomial Theorem, prove:
$$ (1+x)^4 + (1-x)^4 = 2(1 + 6x^2 + x^4) $$
Solution:
Expand $(1+x)^4$:
$$ 1 + 4x + 6x^2 + 4x^3 + x^4 $$
Expand $(1-x)^4$:
$$ 1 - 4x + 6x^2 - 4x^3 + x^4 $$
Add:
$$ (1+x)^4 + (1-x)^4 = (1+1) + (4x - 4x) + (6x^2 + 6x^2) + (4x^3 - 4x^3) + (x^4 + x^4) $$
$$ = 2 + 0 + 12x^2 + 0 + 2x^4 = 2(1 + 6x^2 + x^4) $$

20. Find the value of x if ratio of 4th to 5th term in $(1+x)^7$ is $2:3$.
Solution:
4th term:
$$T_4 = {}_7C_3 x^3 = 35x^3$$
5th term:
$$T_5 = {}_7C_4 x^4 = 35x^4$$
Given:
$$\frac{T_4}{T_5} = \frac{2}{3}$$
Substitute:
$$\frac{35x^3}{35x^4} = \frac{2}{3} \implies \frac{1}{x} = \frac{2}{3} \implies x = \frac{3}{2}$$